

Platforms as Search Technologies

Price timing, seller participation, and competition

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Prices are visible before a traveler knows which hotel fits best

A hotel-booking example

A traveler sees room prices on hotel websites and a booking platform. She still needs to inspect locations, room features, photos, and reviews to learn which hotel best fits her trip. Direct search means inspecting hotels one by one; the platform compares many at once.

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Posted prices $\implies$ Where to search $\implies$ Hotel fit

Posted prices can affect the consumer's **search channel**.

Armstrong–Zhou (2011); Haan–Moraga-González–Petrikaitė (2018); Ding–Zhang (2018); Choi–Dai–Kim (2018)

A hotel's direct price affects both search and checkout

Low direct price



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Before search

Direct search looks attractive

Fewer buyers use the platform

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After the platform identifies a hotel

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The purchase is completed directly

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Showrooming

The same low price can keep the traveler off the platform before search or pull her away after discovery.

Prices affect the size of the platform

q_j : hotel j 's direct price; p_j : its platform price; λ : share using the platform.

Current prices are observed first

$$\frac{\partial \lambda}{\partial q_j} > 0, \quad \frac{\partial \lambda}{\partial p_j} < 0$$

Hotel j can influence where consumers begin searching through its platform and **direct** prices.

Search-steering timing: this paper; related price-directed-search models.

The channel is chosen from expectations

$$\frac{\partial \lambda}{\partial q_j} = 0, \quad \frac{\partial \lambda}{\partial p_j} = 0$$

Expected prices can affect platform use, but one seller's realized price cannot redirect initial search.

Wang–Wright (2025); models that begin after a platform lead already exists.

Three questions

Question 1

How do prices steer search and checkout?

Policy question: When does a commission cap help consumers?

Main part of the talk

Question 2 (companion result)

When sellers choose whether to join, how does the fee structure shape the assortment?

Policy: If a commission cap leads to higher fixed fees, can it hurt consumers?

Question 3

How does ownership change incentives to steer?

Outlook

We first hold the seller set fixed, then study how fees select sellers (shutting down price steering), and finally give a brief outlook on platform competition.

Products and information

- ▶ $J \geq 2$ sellers
- ▶ consumer i 's value for j is

$$V_{ij} = x_{ij} + z_{ij}$$

- ▶ x_{ij} becomes known after channel choice; z_{ij} is learned by inspection
- ▶ x_{ij} and z_{ij} are independent continuous draws

Demand (Cho-Dai-Kim) and Prices

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Prices and fees

- ▶ direct price q_j ; platform price p_j
- ▶ commission τ per platform sale
- ▶ buyer fee t for using the platform
- ▶ switching cost κ_i : independent continuous draw

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Timing: fees $\rightarrow$ posted prices $\rightarrow$ search channel $\rightarrow$ product information $\rightarrow$ checkout.

Platform versus Direct Search

All p_j and q_j are observed before search. Match value z_{ij} is not.

Direct search (as in Cho-Dai-Kim)

Inspect sellers one by one at cost s . The reservation value z_j^* solves

$$s = \mathbb{E}[(z_{ij} - z^*)_+].$$

Platform search (similar to Bergemann-Bonatti)

The platform ranks offers by net utility:

$$\max_j \{x_{ij} + z_{ij} - p_j\}.$$

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The platform's advantage is informational, reduces inspection cost to zero.

Weitzman (1979); Choi-Dai-Kim (2018)

Lemma 1: Platform search supports a higher stand-alone price

Set platform fees to zero and consider each search channel separately.

Price comparison

Under the observable-price search conditions of Choi–Dai–Kim,

$$p_{\text{platform}}^* \geq q_{\text{direct}}^*.$$

The inequality is strict when costly direct inspection binds.

Direct sellers use a low posted price to attract inspection. The platform removes that incentive.

Regularity: log-concavity of the relevant density, cdf, and survival function, with unbounded upper support.

Choi–Dai–Kim (2018), Propositions 3 and 5

Value of Platform (Platform Affinity)

Consumers differ in their value of using the platform:

$$\eta_i \sim R_\eta, \quad r_\eta > 0 \text{ is its density.}$$

Let $U^D(q)$ and $U^P(p)$ be expected utility from direct and platform search before product fit is learned. Consumer i uses the platform iff

$$\eta_i - t + U^P(p) \geq U^D(q).$$

Therefore, platform adoption is

$$\lambda = 1 - R_\eta(t + U^D(q) - U^P(p)).$$

We later use uniform affinity for a closed-form illustration.

Benchmark 1: Search steering without showrooming

Set $\kappa = \infty$. A consumer who uses the platform must also complete the purchase there.

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Both prices affect whether consumers search directly or use the platform.

Lemma 2: Sellers raise the direct price and lower the platform price

Denote seller j 's profit per direct and platform user

$$m_j^D = (q_j - c)D_j^D, \quad m_j^P = (p_j - c - \tau)D_j^P.$$

Search-steering wedges

Assume adoption and prices are interior, demands are differentiable, and each channel-profit function is strictly single-peaked. In Benchmark 1,

$$\frac{\partial \lambda}{\partial q_j} = r_\eta (t + U^D - U^P) D_j^D > 0, \quad \frac{\partial \lambda}{\partial p_j} = -r_\eta (t + U^D - U^P) D_j^P < 0.$$

If $m_j^P > m_j^D$, then in a symmetric equilibrium: $q_j > q_{\text{direct}}^*$, $p_j < p_{\text{platform}}^*$.

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Under covered demand, $D_j^P = D_j^D = 1/J$, so the condition reduces to $p_j - c - \tau > q_j - c$. Lemma 1 (Choi–Dai–Kim) show that eliminating search costs raises sellers' prices and profits.

Commission Cap

A lower commission raises the seller's margin from an on-platform sale:

$$p_j - c - \tau \quad \uparrow \quad \text{when } \tau \downarrow .$$

When platform customers are relatively more profitable, the seller wants more consumers to search there.

Lower the platform price

$$p_j \downarrow$$

Platform search becomes more attractive.

Raise the direct price

$$q_j \uparrow$$

Direct search becomes less attractive.

A commission cap affects consumers through prices and platform use

Commission $\tau \implies$ Prices $p, q \implies$ Platform share $\lambda \implies$ Consumer surplus

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The last step has two parts. Under covered demand, along a differentiable equilibrium path without showrooming ($\kappa = \infty$),

$$\frac{dCS}{d\lambda} = \underbrace{\frac{\lambda}{r_{\eta}(R_{\eta}^{-1}(1 - \lambda))}}_{\text{utility gain from platform}} - \underbrace{\frac{dq}{d\lambda}}_{\text{direct-price effect}}.$$

Platform utility

More consumers get utility from using the platform.

Direct-price loss

Consumers who remain direct pay more.

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To evaluate a cap, we need: how the fee changes q , and what the marginal utility gains from the platform are.

Wang–Wright benchmark: No search steering

Timing

1. The platform sets commission τ .
2. Sellers choose prices while consumers choose a channel from expected equilibrium prices.
3. Consumers then search and buy within the chosen channel.

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Pass-through with inelastic demand

Let μ_D and μ_P be the equilibrium supplier markups in the direct and platform channels. Then

$$q^* = \mu_D, \quad p^* = \mu_P + \tau,$$

$$\tau \downarrow \implies p \downarrow \text{ one-for-one, } q \text{ unchanged.}$$

Direct price cannot steer search to a different channel, so the cap generates no direct-price response.

Two benchmarks

Prices observed before channel choice

A lower commission makes platform sales relatively more attractive.

The seller lowers p **and raises** q to steer search.

Consumer surplus can fall.

Wang–Wright timing

Consumers react to expected equilibrium prices, not one realized price.

The platform price falls; the direct price is unchanged.

In the baseline, consumer surplus rises.

The extra loss with search steering is borne by consumers who continue to buy directly.

Unified framework: Search first, checkout second

Initial search

Direct or platform?

Platform share: λ



Checkout

Platform or direct?

External share: $\varphi =$
 $Pr(\text{direct checkout} | \text{platform search})$

At symmetric prices, a platform user leaves when the direct-price saving covers the switching cost:

$$\kappa_i \leq p - q.$$

Let φ be the share who leave. Then

$$\omega \equiv \lambda(1 - \varphi)$$

is the share of all consumers who both use and complete on-platform.

Policy incidence depends on where platform users finally buy.

Who gains from a lower commission?

Suppose a commission cut lowers the platform price and raises the direct price, i.e.,

$$\frac{dp}{d\tau} > 0, \quad \frac{dq}{d\tau} < 0.$$

With covered demand,

$$\frac{dCS}{d\tau} = -\omega \frac{dp}{d\tau} - (1 - \omega) \frac{dq}{d\tau}.$$

A commission cut helps consumers iff enough purchases are completed on-platform relative to the two price responses.

Who gains from a lower commission?

With covered demand, $\eta \sim U[\underline{\eta}, \bar{\eta}]$. At every sufficiently low-adoption equilibrium,

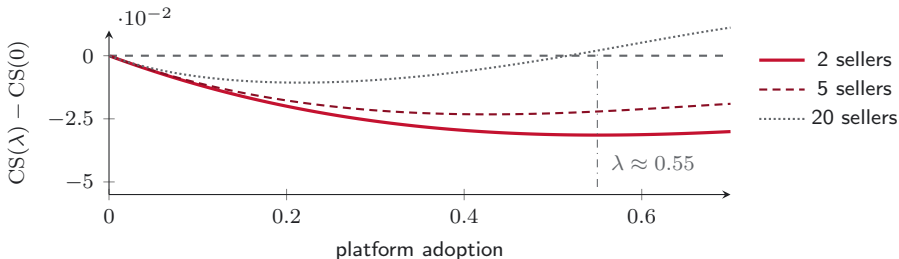
$$\tau \downarrow \implies p \downarrow, \quad q \uparrow, \quad \lambda \uparrow, \quad \varphi \downarrow \text{ (less showrooming),} \quad \text{CS} \downarrow, \quad \text{TS} \uparrow.$$

- ▶ **Consumer surplus falls:** most consumers still purchase directly, so the increase in q reaches more buyers than the reduction in p .
- ▶ **Total surplus rises here:** at low adoption, extra platform search improves matching, while less showrooming saves switching costs. Prices and platform fees are transfers.

This is a local comparative statics at symmetric equilibria.

Consumer harm need not be confined to a tiny platform

Uniform affinity with support width $\Delta_\eta = 0.25$; the platform markup advantage is 0.15.



With two sellers, consumer surplus falls until $\lambda \approx 0.55$. More sellers weaken each seller's steering power and lower the direct price effect.

A tighter cap raises λ ; the downward-sloping part of each curve is where the cap harms consumers.

When is a commission cap more likely to help?

More likely to help

Most platform users complete there

Many sellers compete for each buyer

Direct prices and buyer fees respond little

Transaction-oriented platforms

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Many platform-generated leads convert directly
A few sellers strongly affect platform use
Direct prices or buyer fees rise substantially

Discovery and comparison platforms

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Discovery and comparison platforms

Measure completion by channel, seller concentration, both prices, and the fee that replaces the capped commission.

NYC DCWP (2026); City of Seattle (2026)

Parity bans

Parity clause forces the platform price down and the direct price up toward one common price $q < r < p$. In the covered uniform- η benchmark, at the same platform fees:

$$CS^{\text{separate}} - CS^{\text{parity}} = \underbrace{(r - q)}_{\text{lower direct price}} - \underbrace{\frac{\bar{\eta} - \eta}{2} [(\lambda^{\text{parity}})^2 - (\lambda^{\text{separate}})^2]}_{\text{lost platform participation rents}}.$$

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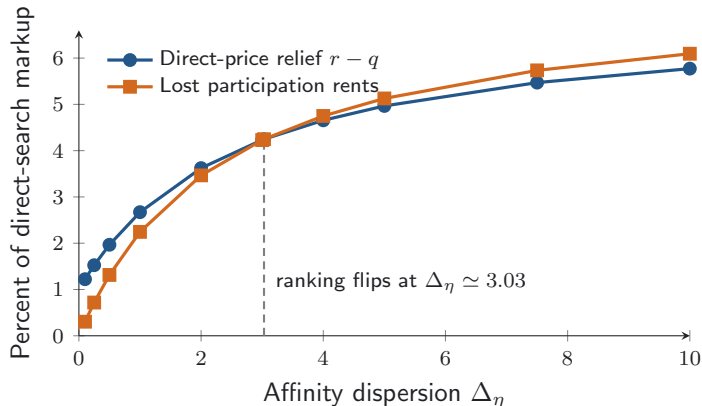
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It substantially reduces platform use, so consumers lose large participation rents from platform search.

A ban helps consumers iff direct-price relief exceeds the lost participation rents. The platform's matching advantage is already reflected in the two equilibrium adoption levels.

Parity or separate prices? An illustration



The ban helps when direct-price relief exceeds the participation rents lost as λ falls.

Calibration

$$J = 2, q = 1, p = 1.15$$

$$c = t = \tau = 0$$

$$K_0 = \Delta_\eta / 2$$

$$\text{so that } \lambda^{\text{par}} = \frac{1}{2}.$$

Consumer ranking

$\Delta_\eta < 3.03$:
separate prices better

$\Delta_\eta > 3.03$:
parity better

Question 2: How is the composition of sellers affected by the fee structure?

The analysis so far holds the seller set fixed. A companion model isolates seller participation, but shuts down search steering and showrooming.

Timing

1. The platform sets (f, τ, t) .
2. Sellers decide whether to list.
3. Consumers observe the assortment $\mathcal{A}$ and choose a channel.
4. Prices and purchases follow.

Seller j 's gain from listing

$$L_j = \underbrace{\lambda_{\mathcal{A}} \pi_j^P}_{\text{profit from platform users}} - \underbrace{\delta_j \pi_j^D}_{\text{direct profit lost when the listing attracts users}}, \text{ where}$$

$$\delta_j = \lambda_{\mathcal{A}} - \lambda_{\mathcal{A} \setminus \{j\}}.$$

Seller j lists iff $L_j \geq f$.

A seller joins if she performs well on-platform or has less direct business to protect.

A common listing fee can remove small sellers

Suppose a commission cap is followed by a higher common listing fee.

Commission cut

- ▶ Lowers prices for a fixed assortment
- ▶ Saves large sellers more because they make more platform sales

Higher listing fee

- ▶ Is the same for every seller
- ▶ Can push low-demand sellers off the platform

Replacement fee	Assortment	Consumers
Small	Expands	Gain
Intermediate	Some small sellers exit	Still gain
Large	More of the tail exits	Lose

Headline logic

More platform use



Stronger incentive to raise the direct price

A merger may raise commissions but reduce platform use, weakening this steering motive.

A merger may raise commissions and lower direct prices. This is an agenda, not yet a theorem.

1. With search steering, a commission cap can lower the platform price while raising the direct price.
2. Consumer incidence depends on the share of consumers completing on-platform and on the two equilibrium price responses.
3. If a platform recovers revenue through a fixed listing fee, it can push small sellers off the platform.

Empirical test: Who receives the lower price, who faces the higher price, and which products disappear?